

MULTI-AGENT AI FOR GAME DEVELOPMENT

ASSIGNMENT #09

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ADVERSARIAL QA AGENT (OPTIONAL)

DUE DATE: 27 August, 11:59 PM ET

TIME ESTIMATE: 4–6 hours

TOTAL POINTS: 10 points

OVERVIEW

Your objective is to build an adversarial testing agent that runs against your capstone game. The agent's goal is to actively try and break the game rather than just play it normally.

⚠ Important Note: If your game isn't testable by an adversarial agent yet, you should spend this time working on your capstone project instead.

CORE REQUIREMENTS

- **Run a Testing Loop:** The agent must run continuously inside your game, cycling through behaviors like movement, interaction, and boundary-probing.
- **Log Bugs & Edge Cases:** Detect and record boundary breaks, stuck states, unintended collisions, exploits, or gameplay logic violations encountered during the loop.
- **Output a Structured Report:** Format the findings in a structured data format (JSON or CSV). Another developer should be able to read this report and act on it immediately. Required fields include:
 - Location
 - Error Type

- Game Context

DELIVERABLES

- 01. Agent Code:** The scripts/code used to run your adversarial tester.
- 02. Structured Report:** A JSON, CSV, or equivalent file containing the results from at least one test run.
- 03. Short ReadMe:** A brief document answering two questions: What did the agent find? Were you surprised by the findings?

GRADING RUBRIC

CRITERION	DESCRIPTION	POINTS
Findings	The agent found at least one real bug, exploit, or edge case. The report specifically names the mechanic or system where it occurred.	/ 4.0
Agent Logic	The agent actively tries to break the game (not just play it) and utilizes a clear strategy for what "broken" means.	/ 3.0
Structured Report	The output is correctly formatted (JSON/CSV) with actionable fields: location, error type, and game context.	/ 2.0
ReadMe	Describes exactly what the agent found and whether those findings were surprising.	/ 1.0

TOTAL SCORE: / 10.0